Year 1 AS Statistics and Mechanics

Chapter 11

Variable acceleration

Practice Booklet

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11.1 Functions of time

Practice Book, printed page 77.

1. A particle P moves in a straight line. At time t seconds ($t \geq 0$), the displacement, s metres, of P from a point O is given by $s = 3t^3 - 2t^2$. Find:

a. s when $t = 2$

b. the time taken for the particle to return to O.

Hint

When the particle returns to O the displacement, $s = 0$. ← Section 9.3

2. A body moves in a straight line such that its velocity, v m s^{-1}, at time t seconds, is given by $v = 3t^2 - 14t + 8$ for $t \geq 0$. Find:

a. the initial velocity

b. the values of t when the body is instantaneously at rest

c. the value of t when the velocity is 13 m s^{-1}

d. the greatest speed of the body in the interval $0 \leq t \leq 3$.

Hint

For part **d**, sketch a velocity–time graph for the motion of the body and use the symmetric properties of the quadratic curve. **Speed** is the magnitude of the velocity, so the greatest speed could occur when the velocity is either positive or negative. ← Pure Year 1, Chapter 2

3. A particle P moves along the x-axis. At time t seconds, the displacement, s metres, is given by $s = 6t^2 - t^3$. Find:

a. the change in displacement between $t = 3$ and $t = 5$

b. the change in the displacement in the fourth second.

Hint

The fourth second is the time between $t = 3$ and $t = 4$.

4. At time $t = 0$, a toy train is at a point P. The train moves from point P in a straight line and then returns to P along the same straight line. Its distance from P, s m, at time t seconds can be modelled by

$$s = \tfrac{1}{4}(5t - t^2)$$

Find:

a. the maximum distance of the train from P

b. the time taken for the toy train to return to P

c. the total distance travelled in this time

d. the values of t for which the model is valid.

5. E A particle P moves along the x-axis. At time t seconds ($t \geq 0$), the displacement, x metres, of P from the origin O is given by $x = 3t(t^2 - 4t + 4)$. Show that P will never move along the negative x-axis. (2 marks)

6. E/P A particle P moves along the x-axis. At time t seconds the velocity of P is v m s^{-1} in the direction of x increasing, where $v = 2t^2 - 13t + 15$, $t \geq 0$. Find:

a. the times when P is instantaneously at rest (3 marks)

b. the times when the velocity is 9 m s^{-1} (3 marks)

c. the greatest speed of P in the interval $0 \leq t \leq 5$. (5 marks)

7. E/P At time $t = 0$, a body moves in a straight horizontal line from a point O and then returns to its starting point. The displacement, s metres, from the point O at time t seconds is given by $s = 5t^2 - t^3$, $0 \leq t \leq T$.

a. Find the change in the displacement in the third second. (3 marks)

b. Given that the model is valid when $s \geq 0$, find the value of T. Justify your answer. (3 marks)

8. E/P At time $t = 0$, a particle P leaves the origin O and moves along the x-axis. At time t seconds the velocity of P is v m s^{-1}, where $v = 10t - t^2$. Find:

a. the times when P is instantaneously at rest (2 marks)

b. the maximum value of v. (4 marks)

11.2 Using differentiation

Practice Book, printed page 78.

1. Find the velocity of each particle at time $t = 2$, given that the particle's displacement (s metres) at time t seconds is given by:

a. $s = 4t^3 - t$

b. $s = 2t + t^2 - 1$

c. $s = t + t^{-1}$

Hint

Differentiate the expression for the displacement to find an expression for the velocity, then substitute $t = 2$ in that expression. The units of velocity will be m s^{-1}.

2. Find an expression for **i** the velocity and **ii** the acceleration of a particle given that the displacement, s metres, at time t seconds is given by:

a. $s = \dfrac{3t^4}{4} - \dfrac{1}{t^2}$

b. $s = (3t - 5)(2t^2 + 1)$

c. $s = \dfrac{2t^4 - 3t^2 + 8}{4t}$

Hint

$v = \dfrac{\mathrm{d}s}{\mathrm{d}t}$, $a = \dfrac{\mathrm{d}v}{\mathrm{d}t} = \dfrac{\mathrm{d}^2s}{\mathrm{d}t^2}$

3. A particle P is moving along the x-axis. At time t seconds the displacement, x metres, from O is given by $x = 2t^3 - 5t^2 + 8$, $t \geq 0$. Find:

a. the velocity of P when $t = 4$

b. the values of t for which P is instantaneously at rest

c. the acceleration of P when $t = 2$.

Hint

The particle is instantaneously at rest when $v = 0$.

4. A particle P is moving along the x-axis. At time t seconds (where $t \geq 0$), the velocity of P is v m s^{-1} in the direction of x increasing, where $v = 20 + t - t^2$. Find the acceleration of P when P is instantaneously at rest.

5. E A particle P moves in a straight line. At time t seconds, the displacement, s cm, from a fixed point O is given by

$$s = \tfrac{1}{3}(2t^3 - 21t^2 + 60t + 90), \ t \geq 0$$

Find:

a. the distance between the points at which the particle is instantaneously at rest (7 marks)

b. the magnitude of the acceleration at these points. (3 marks)

6. E/P A particle P moves along the x-axis. At time t seconds the velocity of P is v m s^{-1} in the direction of x increasing, where $v = 2t^2 - 17t + 21$, $t \geq 0$. Find:

a. the times when P is instantaneously at rest (3 marks)

b. the greatest speed of P in the interval $0 \leq t \leq 6$. (5 marks)

7. E/P A particle P moves in a straight line. At time t seconds the displacement, s metres, is given by $s = kt^3 - 5t^2 + 12$, where k is a constant. When $t = 1$, the velocity of P is 8 m s^{-1}.

a. Find the value of k. (3 marks)

b. Hence find the magnitude of the acceleration when P is instantaneously at rest. (5 marks)

8. E/P A particle P moves along a straight line. At time t seconds the velocity of P is v m s^{-1}, where

$$v = \tfrac{1}{10}t(8 - t)^2, \ 0 \leq t \leq 10$$

a. Sketch a velocity–time graph for the motion of P. (2 marks)

b. Find the values of t and the corresponding values of v when the acceleration of P is instantaneously zero. (5 marks)

11.3 Maxima and minima problems

Practice Book, printed page 79.

1. A particle starts at a fixed point O and moves in a straight line before returning to O. The distance, s metres, from O at time t seconds is given by $s = 3t + 0.25t^2 - 0.25t^3$, $0 \leq t \leq 4$.

a. Justify the restriction $0 \leq t \leq 4$.

b. Find the maximum distance of the particle from O, correct to 3 significant figures.

Hint

For part **a**, find the roots of the cubic function and draw a sketch. ← Pure Year 1, Section 4.1
For part **b**, look at your sketch. If the maximum distance occurs at a turning point, you can find the value of t by solving $\frac{ds}{dt} = 0$. ← Pure Year 1, Section 12.9

2. At time $t = 0$, a particle P leaves the origin, O, and moves along the x-axis. At time t seconds the velocity of P is v m s^{-1} where $v = t^2(t-5)^2$, $0 \leq t \leq 6$.

a. Sketch a velocity–time graph for the motion of P.

b. Find the maximum value of v and the time at which it occurs.

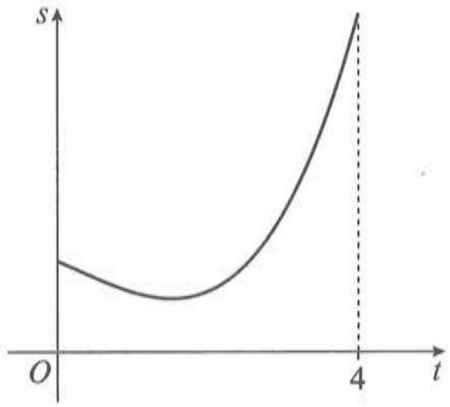
Hint

Set $\frac{dv}{dt} = 0$, or use the symmetry of the graph. ← Pure Year 1, Section 12.9

3. A particle P moves in a straight line such that its distance, s metres, from a fixed point O at time t seconds is given by

$$s = 0.5t^3 - 0.45t^2 - 2.1t + 6, \ 0 \leq t \leq 4$$

The diagram shows the displacement–time graph of the motion of P.



a. Determine the time at which P is moving with minimum velocity.

b. Find the displacement of P from O at this time.

c. Find the velocity of P at this time.

4. A body starts from rest and moves in a straight line. At time t seconds, the displacement of the body from the starting point, s metres, is given by $s = 5t^3 - t^4$, $0 \leq t \leq 5$.

a. Find the time taken for the particle to return to the starting point.

b. Explain why s is always non-negative.

c. Find the maximum displacement of the body from its starting point.

5. E A particle P moves in a straight line such that its displacement, s metres, from a fixed point O at time t seconds is given by $s = 0.5t^3 - 0.3t^2 - 0.8t + 6$, $0 \leq t \leq 4$. Find the displacement of P from O when it is moving with minimum velocity. (6 marks)

6. E/P A particle P moves along the x-axis. Its velocity, v m s^{-1}, in the positive x-direction at time t seconds is given by $v = 3t^2 - 6t + 8$, $t \geq 0$.

a. Show that P never comes to rest. (2 marks)

b. Find the minimum velocity of P. (4 marks)

7. E/P An electric car moves in a straight line along a 5 km test track. The motion of the car is modelled as a particle travelling in a straight line and the distance, s m, of the car from the start of the track after t seconds is given by $s = 4.8t + 1.2t^2 - 0.01t^3$, $0 \leq t \leq 100$. Show that the car never reaches the end of the track. (7 marks)

8. E/P A particle P moves in a straight line. The speed of P at time t seconds ($t \geq 0$) is v m s^{-1}, where $v = at^2 + bt + c$ and a, b and c are constants. When $t = 3$, the speed of P has its minimum value. When $t = 0$, $v = 28$ and when $t = 3$, $v = 1$.
Find the acceleration of P when $t = 4$. (8 marks)

11.4 Using integration

Practice Book, printed page 81.

1. A particle is moving in a straight line. Given that $v = 0$ when $t = 0$, find an expression for the velocity of the particle if the acceleration is given by:

a. $a = 4t^2 - 3t^3$

b. $a = 5t + \dfrac{3t^2}{2}$

c. $a = \dfrac{2t^3 - t^2}{t}$

Hint

$v = \int a \, \mathrm{d}t$. Use the initial condition that $v = 0$ when $t = 0$ to find the constant of integration.

2. A particle is moving in a straight line. Find:

i. an expression for the displacement of the particle

ii. the distance travelled by the particle in the first 3 seconds

when the velocity is given by:

a. $v = 2t^3 - \dfrac{t}{3} + 1$

b. $v = \dfrac{t}{3}(2t^2 + t^3)$

c. $v = 5t^2 + \dfrac{3\sqrt{t}}{2}$

Hint

If the velocity is always positive in the interval $0 \leq t \leq 3$, then the distance travelled is given by the definite integral $\int_0^3 v \, \mathrm{d}t$.

3. A particle P is moving on the x-axis. At time t seconds, the velocity of P is $(10 + 3t - 2t^2)$ m s^{-1}. At time $t = 0$, P is at the point where $x = 5$. Find the distance of P from O when $t = 2$.

4. A particle is moving in a straight line. At time t seconds its velocity, v m s^{-1}, is given by $v = 4t^2 - 20t + 21$. When $t = 0$, the displacement is 0. Find the distance between the two points where the particle is instantaneously at rest.

5. E A particle P moves on the x-axis. The acceleration of P at time t seconds is $(t-3)$ m s^{-2} in the positive x-direction. The velocity of P at time t seconds is v m s^{-1}. When $t = 0$, $v = 4$. Find:

a. v in terms of t (4 marks)

b. the values of t when P is instantaneously at rest (3 marks)

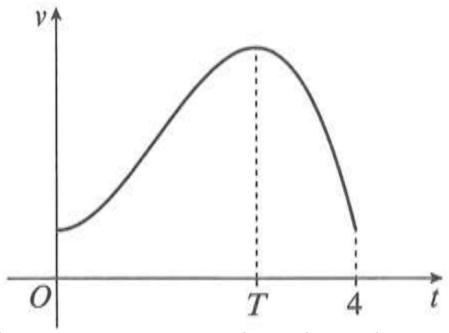
c. the distance between the two points at which P is instantaneously at rest. (4 marks)

6. E/P A particle P moves along the x-axis. At time t seconds ($t \geq 0$), the acceleration of P is $(5t + 9)$ m s^{-2} in the positive x-direction. When $t = 0$, the velocity of P is 3 m s^{-1} in the positive x-direction. At time $t = T$, the velocity of P is 7 m s^{-1} in the positive x-direction. Find the value of T. (6 marks)

7. E/P A particle P starts from rest at a fixed point O and moves in a straight line. At time t seconds the acceleration of P is proportional to $(5 - t^2)$, where $t \geq 0$, and the displacement of P from O is s metres. When $t = 2$, the velocity of P is 11 m s^{-1}.

Show that $s = \dfrac{t^2(30 - t^2)}{8}$. (7 marks)

8. E/P A particle P moves in a straight line through a fixed point, O. The velocity of the particle, v m s^{-1} at time t seconds after passing through O is given by $v = 5 + 8t^2 - 2t^3$, $0 \leq t \leq 4$. The diagram shows a velocity–time graph of the motion of P.



Find the distance of P from O at time T seconds, when the particle is moving with maximum velocity. (7 marks)

11.5 Constant acceleration formulae

Practice Book, printed page 82.

1. A particle moves in a straight line such that its displacement, s metres, from a fixed point O at time t seconds is given by $s = ut + \frac{1}{2}at^2$, where u and a are constants. Use calculus to:

a. find an expression for the velocity v in terms of u, a and t

b. show that the acceleration is a m s^{-2}.

Hint

For part **a**, use $v = \dfrac{\mathrm{d}s}{\mathrm{d}t}$.

For part **b**, use acceleration $= \dfrac{\mathrm{d}v}{\mathrm{d}t} = \dfrac{\mathrm{d}^2s}{\mathrm{d}t^2}$.

2. A particle moves in a straight line with constant acceleration a m s^{-2}. Given that when $t = 0$, the velocity is u m s^{-1} and the displacement is 0 m, find expressions in terms of u, a and t for:

a. the velocity, v m s^{-1}

b. the displacement, s m.

Hint

For part **a**, use $v = \int a \, \mathrm{d}t$.
For part **b**, use $s = \int v \, \mathrm{d}t$.

3. Which of these equations for displacement describe constant acceleration? Explain your answers.

a. $s = 3t - 2t^3$

b. $s = t^2 - 7t$

c. $s = \dfrac{5}{t} + 8$

d. $s = 16 - \dfrac{t^2}{2}$

e. $s = 10$

Hint

Check whether $\dfrac{\mathrm{d}^2 s}{\mathrm{d}t^2}$ is a constant.

4. E A car travels along a straight road. It passes through point A when $t = 0$ and through point B 50 seconds later. Its distance from A at time t seconds is given by $s = 32t - 0.3t^2$, $0 \leq t \leq 50$.

a. Find the distance AB. (1 mark)

b. Show that the car travels with constant acceleration. (3 marks)

5. E/P A particle moves in a straight line with a constant acceleration of 3 m s^{-2}.

a. Given that its initial velocity is 10 m s^{-1}, use calculus to show that its velocity at time t seconds is given by $v = 10 + 3t$. (2 marks)

b. Given that the initial displacement of the particle is 5 m, show that $s = 10t + 1.5t^2 + 5$. (3 marks)

6. E/P A particle moves in a straight line such that its displacement, s metres, from a fixed point O at time t seconds is given by $s = pt^2 + qt + r$, where p, q and r are constants.
Given that when $t = 2$, $s = 8$, $v = 7$ and $a = 6$, find the values of p, q and r. (5 marks)

7. E/P A particle moves in a straight line with constant acceleration. The initial displacement of the particle is 3 m and the initial velocity of the particle is 5 m s^{-1}. After 2 seconds it is moving with velocity 13 m s^{-1}. Find:

a. the acceleration of the particle (3 marks)

b. an equation for the displacement of the particle, s metres, in terms of t. (5 marks)

Problem solving: Set A

Bronze

A particle P moves along the x-axis. At time t seconds ($t \geq 0$), the displacement, x metres, of P from the origin O is given by $x = t^3 - 8t^2 + 5t$. Find:

a. the times when P is instantaneously at rest (3 marks)

b. the distance between the two points at which P is instantaneously at rest (4 marks)

c. the values of t for which the acceleration is positive. (2 marks)

Silver

A particle P moves along the x-axis. At time t seconds ($t \geq 0$), the displacement, x metres, of P from the origin O is given by $x = t(t^2 - 9t + 15)$. Find:

a. the distance travelled by P in the negative x-direction (6 marks)

b. the total distance travelled by P on the negative x-axis. (2 marks)

Gold

A particle P moves along the x-axis. At time t seconds ($t \geq 0$), the displacement, x metres, of P from the origin O is given by $x = \frac{1}{3}t^2(t^2 - 6t + 10)$.

a. Show that P will never move along the negative x-axis. (2 marks)

Paula substitutes $t = 3$ into the expression for x and obtains an answer of 3. She states that the particle has moved a total distance of 3 m in the first three seconds of its motion.

b.

i. Explain the mistake Paula has made.

ii. Determine the distance the particle travels in the first three seconds of its motion.

(7 marks)

Problem solving: Set B

Bronze

A particle starts from a point, P, and moves in a straight line. At time t seconds after the beginning of its motion, the acceleration of the particle, a m s^{-2}, is given by $a = 5 - 2t$ for $t \geq 0$. When $t = 0$, $v = 6$ m s^{-2}. Find:

a. the time when the particle is instantaneously at rest (6 marks)

b. the displacement of the particle from P at this time. (3 marks)

Silver

A particle starts from rest and moves in a straight line from a point, P. At time t seconds after the beginning of its motion, the acceleration of the particle, a m s^{-2}, is given by $a = 9 - 6t$ for $0 \leq t \leq 3$. Find:

a. the greatest speed of the particle (6 marks)

b. the total distance travelled by the particle in the interval $0 \leq t \leq 3$. (4 marks)

Gold

A particle starts from rest and moves in a straight line from a point, P. At time t seconds after the beginning of its motion, the acceleration of the particle, a m s^{-2}, is given by $a = 26 - 12t$. Find:

a. the total distance travelled by the particle in the first 5 seconds (7 marks)

b. the time taken for the particle to return to P. (5 marks)

Selected exam question bank questions

Question 46

A particle P moves along the x-axis. At time t seconds the velocity of P is v m s^{-1} in the positive x-direction, where $v = 2t^2 - 3t + 4$. When $t = 0$, P is at the origin O. Find the distance of P from O when P is moving with minimum velocity. (8 marks)

Question 49

A particle P moves along a straight line. The speed of P at time t seconds ($t \geq 0$) is v m s^{-1}, where $v = (pt^2 + qt + r)$ and p, q and r are constants. When $t = 3$, the speed of P has its minimum value. When $t = 0$, $v = 10$ and when $t = 2$, $v = 5$. Find:

a. the acceleration of P when $t = 4$ (8 marks)

b. the distance travelled by P in the third second of the motion. (5 marks)

Question 50

A car travels in a straight line. It passes through a point A when $t = 0$ and through a point B when $t = 30$. Its distance from A at time t seconds is given by $s = 25t - 0.3t^2$, $0 \leq t \leq 30$.

A second car travels in the opposite direction along the same road as the first car. It passes through B when $t = 0$ with an initial velocity of 15 m s^{-1} and travels directly towards A with a constant acceleration of 0.1 m s^{-2}. Find:

a. the distance AB (2 marks)

b. the distance from A at which the two cars pass one another. (8 marks)

Question 54

A particle moves in a straight line. At time t seconds, the particle's displacement, s m, is given as $s = (2 - t)^4$, $t \geq 0$. Show that:

a. $v \geq 0$ when $t \geq k$ (where v is the particle's velocity in m s^{-1}) and state the least possible value of k. (5 marks)

b. the acceleration, a m s^{-2}, can be written as $a = 12(t - 2)^2$. (4 marks)

Question 56

A particle moves in a straight line. At time t seconds, the acceleration of the particle, a m s^{-2}, is given by $a = 3 - \dfrac{2}{t^3}$.

When $t = 1$, the particle is at rest at a fixed point P on the line. Find, in terms of t:

a. an expression for the velocity of the particle (5 marks)

b. an expression for the displacement, s m, of the particle from P. (5 marks)

Question 58

A particle moves in a straight line with acceleration a m s^{-2}. At any time, t seconds, a is inversely proportional to t^3. When $t = 1$, the velocity of the particle is 2 m s^{-1}. Given the velocity approaches a limiting value of 6 m s^{-1}, find an expression for the velocity at any time t. (9 marks)

Answers

11.1 Functions of time

1.

a. 16

b. $\frac{2}{3}$ seconds

2.

a. 8 m s^{-1}

b. $\frac{2}{3}$ seconds and 4 seconds

c. 5 seconds

d. $\frac{25}{3}$ m s^{-1}

3.

a. -2 m

b. 5 m

4.

a. $\frac{25}{16}$ m

b. 5 seconds

c. $\frac{25}{8}$ m

d. $0 \leq t \leq 5$

5. $x = 3t(t-2)^2 \geq 0$ for all $t \geq 0$.

6.

a. $t = 1.5$ and $t = 5$ seconds

b. $t = 0.5$ and $t = 6$ seconds

c. 15 m s^{-1}

7.

a. 6 m

b. $5t^2 - t^3 = t^2(5-t) \geq 0$ when $t \geq 0$ and $t \leq 5$, so $T = 5$

8.

a. $t = 0$ and $t = 10$ seconds

b. 25 m s^{-1}

11.2 Using differentiation

1.

a. 47 m s^{-1}

b. 6 m s^{-1}

c. 0.75 m s^{-1}

2.

a.

i. $3t^3 + \frac{2}{t^3}$

ii. $9t^2 - \frac{6}{t^4}$

b.

i. $18t^2 - 20t + 3$

ii. $36t - 20$

c.

i. $\frac{3t^2}{2} - \frac{3}{4} - \frac{2}{t^2}$

ii. $3t + \frac{4}{t^3}$

3.

a. 56 m s^{-1}

b. $t = 0$ and $t = \frac{5}{3}$

c. 14 m s^{-2}

4. -9 m s^{-2}

5.

a. 9 cm

b. 6 cm s^{-2}

6.

a. $t = 1.5$ and $t = 7$ seconds

b. Greatest speed = 21 m s^{-1} (when $a = 0$, $v = -15.125$)

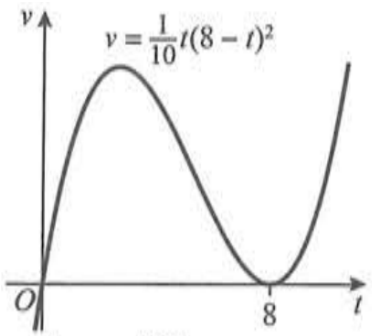
7.

a. $k = 6$

b. 10 m s^{-2}

8.

a.



b. $t = \frac{8}{3}$ s, $v = \frac{1024}{135}$ m s^{-1} and $t = 8$ s, $v = 0$ m s^{-1}

11.3 Maxima and minima problems

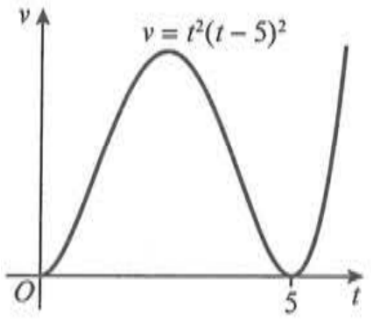
1.

a. $s = 0$ at $t = 0$ and $t = 4$. s represents distance so must be positive. $s \geq 0$ when $0 \leq t \leq 4$.

b. 5.19 m

2.

a.



b. $v = 39.1$ m s^{-1} when $t = 2.5$ seconds

3.

a. 0.3 seconds

b. 5.343 m

c. -2.235 m s^{-1}

4.

a. 5 seconds

b. $s = t^3(5 - t)$
For $0 \leq t \leq 5$, $t^3 \geq 0$ and $5 - t \geq 0$ so $s \geq 0$.

c. $s = 65.9$ m (3 s.f.)

5. 5.832 m

6.

a. $b^2 - 4ac = 36 - 96 = -60 < 0$, so no real roots.

b. 5 m s^{-1}

7. Max distance is when
$\frac{\mathrm{d}s}{\mathrm{d}t} = 4.8 + 2.4t - 0.03t^2 = 0 \Rightarrow t = 82.0$ (3 s.f.).
Max distance = 2.95 km (3 s.f.) so the train never reaches the end of the track.

8. 6 m s^{-2}

11.4 Using integration

1.

a. $v = \frac{4t^3}{3} - \frac{3t^4}{4}$

b. $v = \frac{5t^2}{2} + \frac{t^3}{2}$

c. $v = \frac{2t^3}{3} - \frac{t^2}{2}$

2.

a.

i. $s = \dfrac{t^4}{2} - \dfrac{t^2}{6} + t + c$

ii. $s = 42$

b.

i. $s = \dfrac{t^4}{6} + \dfrac{t^5}{15} + c$

ii. $s = 29.7$

c.

i. $s = \dfrac{5t^3}{3} + t^{\frac{3}{2}} + c$

ii. $s = 50.2$ (3 s.f.)

3. $\frac{77}{3}$ m

4. $\frac{16}{3}$ m

5.

a. $v = \dfrac{t^2}{2} - 3t + 4$

b. $t = 2$ and $t = 4$

c. $\frac{2}{3}$ m

6. $T = \frac{2}{5}$

7. Let $a = k(5 - t^2)$. Integrate this and use $v = 0$ when $t = 0$ and $v = 11$ when $t = 2$ to find k. Integrate again to find s and use $s = 0$ when $t = 0$.

8. 38.6 m (3 s.f.)

11.5 Constant acceleration formulae

1.

a. $v = \dfrac{\mathrm{d}s}{\mathrm{d}t} = u + at$

b. Acceleration $= \dfrac{\mathrm{d}v}{\mathrm{d}t} = 0 + a = a$

2.

a. $v = \int a \,\mathrm{d}t = at + c$. When $t = 0$, $v = u$, so $v = u + at$.

b. $s = \int v \,\mathrm{d}t = ut + \frac{1}{2}at^2$

3. **b**: $a = 2$; **d**: $a = -1$; **e**: $v = 0$ (particle stationary)

4.

a. $AB = 850$ m

b. Acceleration $= \dfrac{\mathrm{d}^2 s}{\mathrm{d}t^2} = \dfrac{\mathrm{d}}{\mathrm{d}t}(32 - 0.6t) = -0.6$ m s^{-2}

5.

a. $v = \int 3 \,\mathrm{d}t = 3t + c$. $v = 10$ when $t = 0 \Rightarrow v = 3t + 10$

b. $s = \int (10 + 3t) \,\mathrm{d}t = 10t + 1.5t^2 + c$
$s = 5$ when $t = 0 \Rightarrow s = 10t + 1.5t^2 + 5$

6. $p = 3$, $q = -5$, $r = 6$

7.

a. $a = 4$ m s^{-2}

b. $s = 3 + 5t + 2t^2$

Problem solving: Set A

Bronze

a. $\frac{1}{3}$ and 5 seconds
b. 50.8 m
c. $t \geq \frac{8}{3}$ s

Silver

a. 32 m
b. 50 m

Gold

a. $t^2 \geq 0$, $t^2 - 6t + 10 = (t-3)^2 + 1 > 0$
b.
  i. The particle is not moving in the same direction for the whole of the motion, so the displacement at $t = 3$ is not the distance the particle moves in the first three seconds of its motion.
  ii. 3.125 m

Problem solving: Set B

Bronze

a. $t = 6$ seconds
b. 54 m

Silver

a. 6.75 m s^{-1}
b. 13.5 m

Gold

a. 87.7 m
b. 6.5 seconds

Selected exam question bank answers

Question 46 2.4375 m

Question 49

a. $\frac{5}{4}$ m s^{-2}
b. $\frac{55}{12}$ m

Question 50

a. 480 m
b. 275.5 m (4 s.f.)

Question 54

a. $s = t^4 - 8t^3 + 24t^2 - 32t + 16$

$$\begin{aligned} v = \frac{\mathrm{d}s}{\mathrm{d}t} &= 4t^3 - 24t^2 + 48t - 32 \\ &= 4(t^3 - 6t^2 + 12t - 8) \\ &= 4(t-2)^3 \geq 0 \text{ for } t \geq 2 \end{aligned}$$

So $k = 2$

b. $a = \dfrac{\mathrm{d}v}{\mathrm{d}t} = 12t^2 - 48t + 48 = 12(t-2)^2$

Question 56

a. $v = 3t + \dfrac{1}{t^2} - 4$

b. $s = \dfrac{3t^2}{2} - \dfrac{1}{t} - 4t + \dfrac{7}{2}$

Question 58 $v = 6 - \dfrac{4}{t^2}$